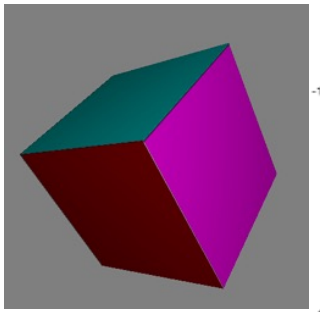
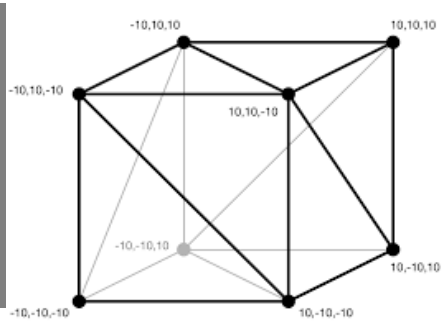
Lab02
Drawing Shapes

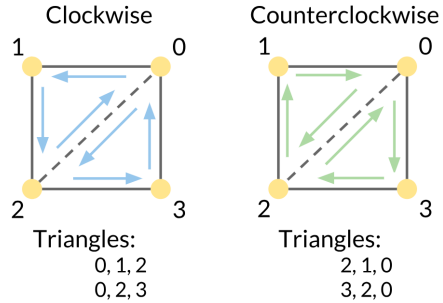
Drawing a Triangular Prism

1. Draw a Triangular Prism using multiple triangles
   - Each Rectangular face is formed by 2 triangles
- Assign a single color for each face as shown in the picture
  - The two triangles on the same rectangular face has the same color

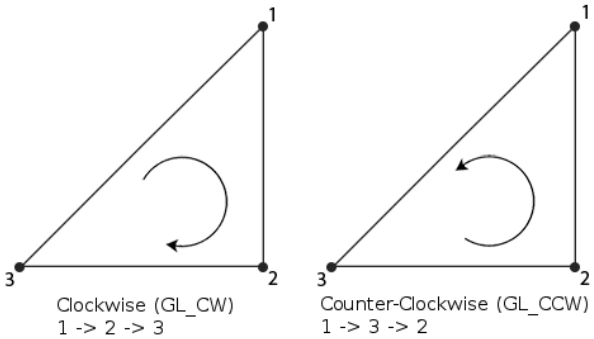




Counter-clockwise



- Pay attention to the order of the vertexes
  - Should be drawn in counter-clockwise order




```
glBegin(GL_TRIANGLE);
glVertex3f(0.5f, 0.5f, 0.0f);
glVertex3f(0.5f, 0.0f, 0.0f);
glVertex3f(0.0f, 0.0f, 0.0f);
glEnd();
```

Display Function

```
void RenderScene(void)
{
   glClearColor(1.0, 1.0, 1.0, 1.0);
   glClear(GL_COLOR_BUFFER_BIT | GL_DEPTH_BUFFER_BIT);
   glMatrixMode(GL_MODELVIEW); // load the modelview matrix
   glLoadIdentity();
   gluLookAt(0,0,10.0f ,0,0,0, 0,1,0);

   //draw
   //use:
   //glColor3f( r, g, b);
   //glBegin(GL_LINE);
   //glVertex3f( -x, 0, 0);
   //glVertex3f( x, 0, 0);
   //glEnd();

   //perform transformation for the cube
   //use:
   //glRotatef(theta, x, y, z);

}
```

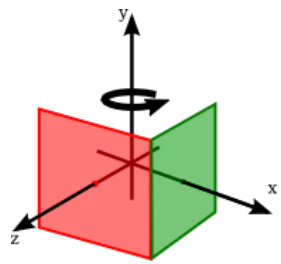
Modify your code here

Rotate your object with this function

Rotate the prism so that we can see at least 3 faces

```
void glRotatef( GLfloat angle,
                GLfloat x,
                GLfloat y,
                GLfloat z);
```

// Specifies the angle of rotation, in degrees.



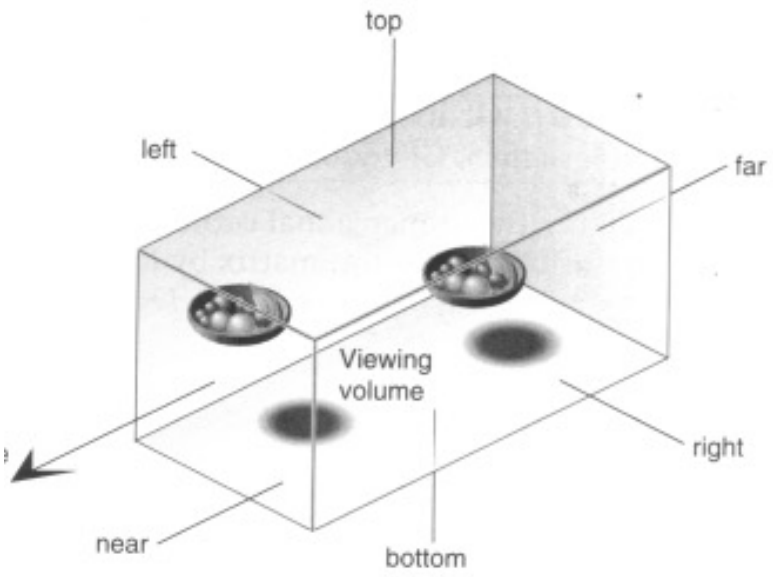
Rotate on Y

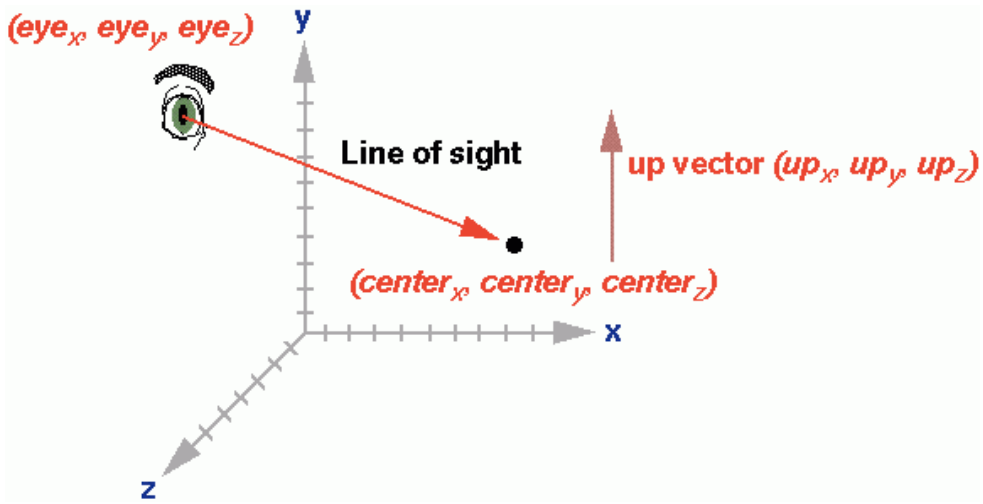
For example:

```
glRotatef(30.0f,1.0,0.0,0.0); // Rotate 30 on x
glRotatef(60.0f,0.0,1.0,0.0); // Rotate 60 on y
glRotatef(90.0f,0.0,0.0,1.0); // Rotate 90 on z
```

glOrtho

- glOrtho(-10,10,-10,10,-10,20);



gluLookAt

- gluLookAt(0,0,-10.0f ,0,0,0, 0,1,0);